Analyzing the Effect of Network Width on Graph Transformers

Auxiliary loss, n = 72, edges in the attention, and a depth limit

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Where we are

- **Task:** for every pair of nodes, are they connected? Output an $n \times n$ yes/no table. `swap` graphs: two blobs, joined or not by two far-apart edges, so no local shortcut works.
- **Width** m: the length of each node's vector. **Critical width** m^*: the smallest m whose test pair accuracy reaches 0.95.
- **Baseline** (adjacency rows, depth 6): $m^* \propto n^{1.74}$ — doubling n needs 3.3× the width.
- **Question since 10-13:** which encodings of the graph make the transformer narrower?

This round: an auxiliary loss, the baseline at $n = 72$, and edges put directly into the attention — which turned out to be limited by **depth**, not width.

Results at a glance

change	m^* at $n = 56$	finding
adjacency rows (baseline)	58	$n^{1.74}$
+ 2-hop rows (input)	18.5	$n^{1.27}$: lower level, slower growth
+ auxiliary k-hop loss	45	$n^{1.30}$: same input, slower growth
edge bias, adjacency tokens	≈ 6	flat over $n = 32$–56
edge bias, random-ID tokens	> 8	rises, unstable

- $n = 72$, baseline: fits training data at the predicted width (81), **limited by data** on test.
- Edge bias past $n \approx 96$: training stalls — **a depth limit**. With 12 layers m^* falls to ≤ 8 / ≈ 8 / ≈ 15 at $n = 72$ / 96 / 128.

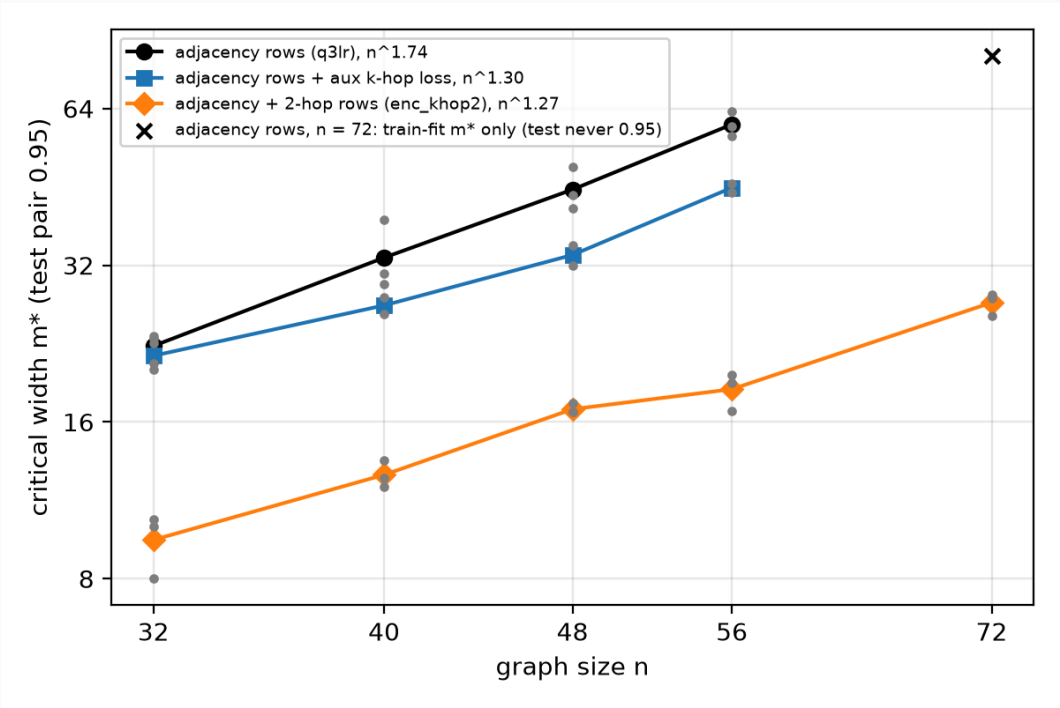
Auxiliary loss and n = 72

A training-only hint changes the exponent

Same input and test-time model. In training, layers 1–3 also predict 2-, 4-, 8-hop reachability (weight 0.3, dropped at test).

n	baseline	+ aux	saving
32	21.9	21.5	2 %
40	30.1	26.8	11 %
48	42.4	33.5	21 %
56	58.0	45.1	22 %
slope	1.74	**1.30**	

m^*, seeds 0 / 1 mean. Slope ranges 1.59–1.90 vs 1.13–1.45: no overlap.



The saving grows with n: the hint changes the **growth**, extra input mostly the **level**.

n = 72: the baseline runs out of data, not width

128k graphs, test (train):

m	seed 0	seed 1
64	.867 (.916)	.898 (.925)
96	.947 (.970)	.912 (.973)
128	.941 (.978)	.936 (.986)

- Train-fit $m^* = 80.9$; the $n = 32$–56 trend predicts 82.6.
- Test never reaches 0.95; train–test gap **0.04–0.08**.

256k graphs, $m = 96$:

graphs	seed 0	seed 1
128k	.947	.912
256k	**.956**	.933

- Gap falls to 0.02, as at $n \leq 56$.
- A test m^* at $n = 72$ needs 256k graphs, more steps, $m = 96$–192.

Edges in the attention

The idea

- Normally a node must **learn** which nodes are its neighbours before it can pass information along edges — that costs width.
- **Edge bias:** every layer adds a learned bonus $b \cdot (A + I)_{ij}$ to the attention score of neighbours. Only direct edges, no distances.
- Each layer then acts like one step of spreading along the edges; after a few layers every node holds a **summary of its component**, and two nodes are connected iff the summaries match.
- Tokens: **adjacency rows** (n wide) or **random IDs** (64 wide, no edges in the input at all).

Pilot at $n = 32$ with 1/8 of the budget: both token types reach **≥ 0.99 at** $m = 16$ (baseline needs 22 with the full budget). A bonus starting at 0 left random IDs stuck.

The needed width stops growing (n = 32–56)

Final test pair accuracy, seed 0 / 1, head dimension 2:

n	$m = 2$	4	6	8
32	.745 / .746	.745 / .746	.991 / 1.00	1.00 / 1.00
40	.750 / .750	.750 / .750	.966 / .993	.994 / .995
48	.755 / .755	.977 / .919	.985 / .985	.988 / .991
56	.759 / .759	.941 / .759	.979 / .884	.971 / .975

Adjacency tokens. 0.75 = always-yes (nothing learned).

n	baseline	edge bias
32	22.4	**5.6**
40	33.1	**5.7**
48	44.7	**4.1**
56	59.6	**5.9**

- Slope ≈ 0: **flat**.
- Random IDs: 4.6, 7.8, > 8, > 8 — rising, non-monotone in m.

But it rises again at n ≥ 72 (depth 6)

Final test pair, seed 0 / 1 (head dim. 8; † = never left always-yes on exact match):

m	$n = 72$	96	128
8	.896 / .909	.831† / .799†	–
16	.987 / .983	.925† / .959	.890† / .855†
24	–	.930 / .948	.916† / .964
32	–	.968 / .968	.930 / .929†

m^*: 13 / 15 at 72; 28 / 16 at 96; > 32 / 23 at 128.

- Train = test everywhere: not data.
- **Not monotone in width:** $n = 128$ seed 1 passes at $m = 24$, fails at $m = 32$.
- Every run: stuck at always-yes 4–7 epochs, then a **second plateau at 0.85–0.93**. Some runs leave it, some don't — that decides m^*, not width.
- Head dim. 2 at $n = 72$ / 96: same rise (11.5 / > 12, > 16 / 15.2) — not the heads.

Why does it stall?

Hypotheses and first checks

Proposed causes (Claude Research report, 11-05)

- **H1** the bonus fades: neighbour share of attention 48 % at $n = 32 \longrightarrow 17$ % at 128
- **H2** escape is luck
- **H3** the learning rate runs out (cosine to 0)
- **H4** unbalanced yes / no pairs
- **H5** too few layers: 6 layers ≈ 12 hops

Data: pairs beyond 12 hops are 0.6 / 3.4 / 9.0 % at $n =$ 72 / 96 / 128 — smooth, **no 12-hop wall**.

Trained models (22 checkpoints):

- b is learned up to 3.3–4, still only 31–49 % neighbour attention at $n = 128$ vs 55–63 % at 72: dilution is real.
- Stalled models err 2–3× more at **every** hop distance.
- Marker of a stalled run: 12–32 % wrong on pairs in different components (6 % escaped).

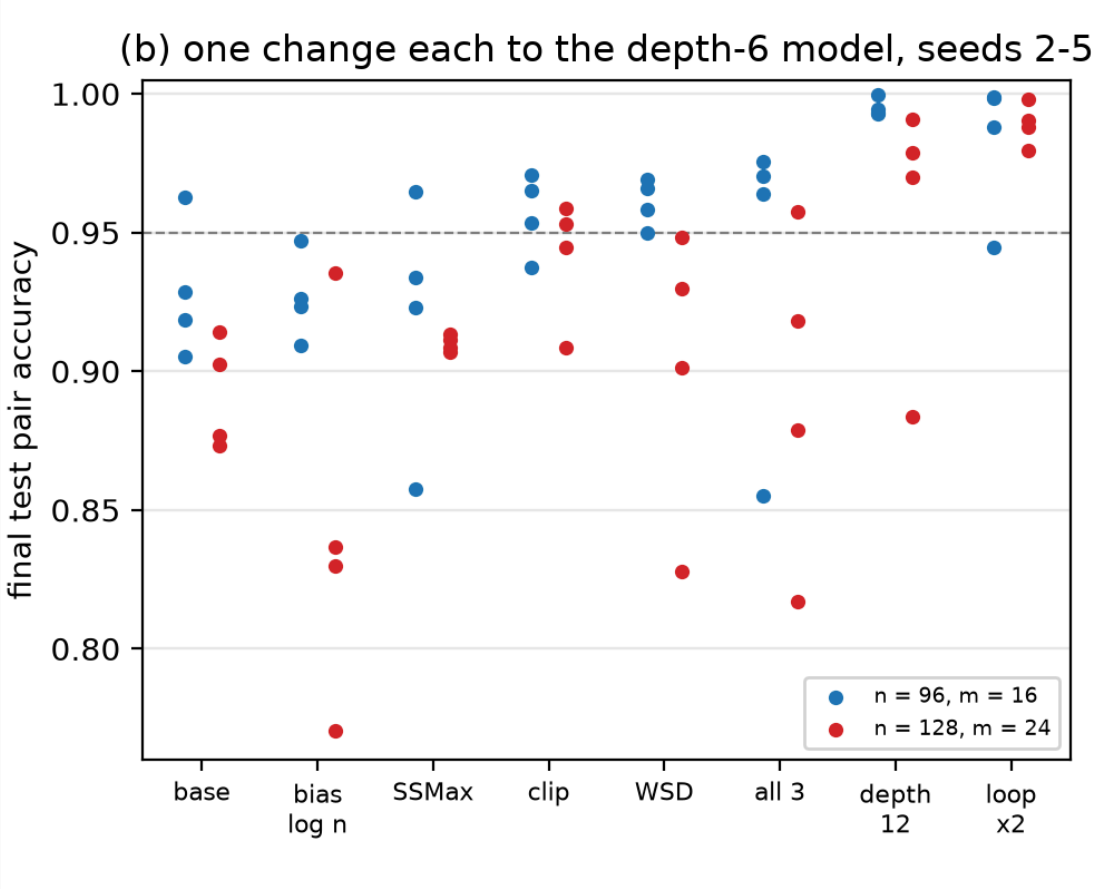
Checkpoint surgery: change one thing, train a bit more

Stalled models resumed for 6000 steps (¼ of the budget). Final test pair:

checkpoint (start)	as is	LR 10^{-3}	$b + 1.5$	+ 6 blocks	blocks twice
$n = 128, m = 16$ (.855)	.857	.857	.855	.980	**.994**
$n = 128, m = 32$ (.930)	.940	.946	.942	.983	**1.000**
$n = 128, m = 32$ (.929)	.932	.951	.932	.995	**1.000**
$n = 96, m = 16$ (.925)	.930	.931	.933	.995	**.998**

More training, a fresh learning rate, a bigger bonus: **stuck**. Running the model's **own 6 layers twice** — no new parameters — **unsticks every one**.

From scratch: one change each, 4 seeds



change	$n = 96$ $m = 16$	$n = 128$ $m = 24$
none	1 / 4	0 / 4
bonus start grows with n	0 / 4	0 / 4
bonus × log n	1 / 4	0 / 4
gradient clipping	3 / 4	2 / 4
WSD schedule	3 / 4	0 / 4
all three	3 / 4	1 / 4
depth 12	4 / 4	3 / 4
6 layers run twice	3 / 4	4 / 4

Seeds reaching test pair 0.95. Plot: final test pair per seed, blue $n = 96$, red $n = 128$.

Conclusion

- The plateau is a **depth limit**. Each layer moves information about one step; past 96 nodes six layers carry it too weakly, at every distance. Width cannot replace the missing layers.
- So the rise of edge-bias m^* past $n \approx 56$ measures **depth**, not width.
- Fits the theory: connectivity needs depth $\Theta(\log n)$ or width $\approx n$.

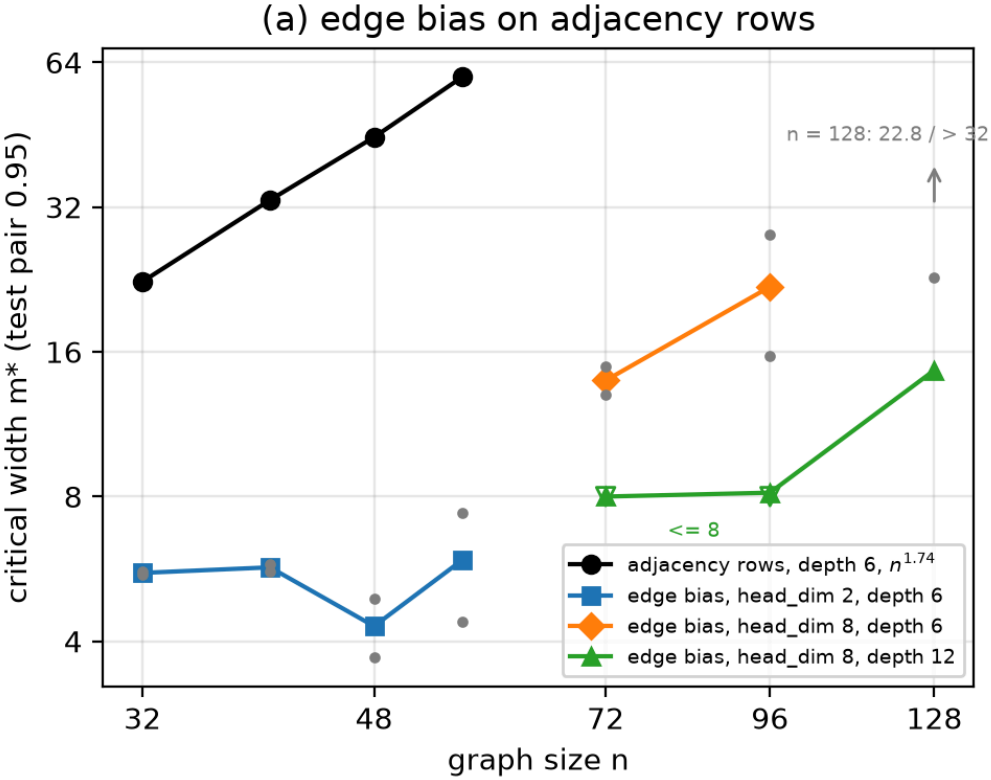
For this encoding, width must be measured with depth tied to n (e.g. $\lceil \log_2 n \rceil$), or as $m^*(n, L)$ on a small depth grid.

Width at depth 12

n	depth 6	depth 12	passing runs
72	13 / 15	≤ 8 / ≤ 8	8 / 8
96	28 / 16	8.3 / ≤ 8	5 / 8
128	> 32 / 23	14.9 / 14.4	3 / 8

m^* seed 0 / 1, widths 8–24, head dim. 8. Passing: runs ≥ 0.95 over all widths.

- **Depth was most of the rise.**
- **Not flat yet** at 128: 12 layers short again, or one 8-wide head at $m \leq 12$?
- **Erratic:** $n = 96$ s1 passes at 8 and 16, 0.815 at 12. Same run twice: .995 / .938.



Next

- Width at a depth that grows with n, **4+ seeds**, readout = fraction passing per width: depth 12 vs 16 at $n = 128$; head dim. 2 at depth 12 for $m^* < 8$.
- Looping (blocks run twice) as cheap depth: best in both ablations, no new parameters.
- Random IDs: test depth 12 / looping — the instability may be the same limit.
- $n = 72$ baseline: 256k graphs, more steps, $m = 96$–192 for a test m^*.
- Thesis ch. 3: add the auxiliary-loss result; update 1.59–1.90 $\longrightarrow$ 1.53–1.97.